

Advanced Java Lab

Week-11

Roll Number: 238W1A12A7

1. Set up H2 in-memory DB with Spring Data JPA and create an entity

Code:

Dependencies:

Spring web, spring data jpa, h2 database

Student.java:

```
package com.example.demo;
```

```
import jakarta.persistence.Entity;
```

```
import jakarta.persistence.Id;
```

```
@Entity
```

```
public class Student {
```

```
    @Id
```

```
    private int id;
```

```
    private String name;
```

```
    public Student() {}
```

```
    public Student(int id, String name) {
```

```
        this.id = id;
```

```
        this.name = name;
```

```
    }
```

```
    public int getId() { return id; }
```

```
public void setId(int id) { this.id = id; }

public String getName() { return name; }

public void setName(String name) { this.name = name; }
}
```

application.properties:

```
spring.h2.console.enabled=true
spring.h2.console.path=/h2-console

spring.datasource.url=jdbc:h2:mem:testdb
spring.datasource.driverClassName=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update
```

2. Create Repository and Demonstrate CRUD

StudentRepository.java:

```
package com.example.demo;

import org.springframework.data.jpa.repository.JpaRepository;

public interface StudentRepository extends JpaRepository<Student, Integer> {
}
```

StudentController.java:

```
package com.example.demo;
```

```
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;
import java.util.List;
```

```
@RestController
@RequestMapping("/students")
public class StudentController {
```

```
    @Autowired
    private StudentRepository repo;
```

```
    // CREATE
    @PostMapping("/add")
    public Student add(@RequestBody Student s) {
        return repo.save(s);
    }
```

```
    // READ all
    @GetMapping("/all")
    public List<Student> getAll() {
        return repo.findAll();
    }
```

```
    // READ by ID
    @GetMapping("/{id}")
    public Student getById(@PathVariable int id) {
        return repo.findById(id).orElse(null);
    }
```

```
// UPDATE

@PutMapping("/update")
public Student update(@RequestBody Student s) {
    return repo.save(s);
}

// DELETE

@DeleteMapping("/delete/{id}")
public String delete(@PathVariable int id) {
    repo.deleteById(id);
    return "Deleted student with id: " + id;
}
}
```

Output:

